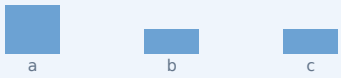
B Local behavior

B1 Tool diversity

$|U(C)|/n$

B2 Command entropy

$-\sum_c p_c \log_2 p_c$



B3 Dominant command

$\max_c p_c$

B4 Action switching

$\sum_t \mathbf{1}[a_t \neq a_{t+1}]/m$

B5 Transition entropy

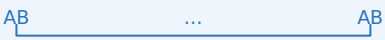
$H(A_{t+1} \mid A_t)$

B6 Longest action run

$L_{\max}/n$

B7 Action-pair reuse

$1 - |U(P_A)|/|P_A|$



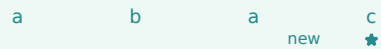
B8 Adjacent repeats

$\sum_t \mathbf{1}[c_t = c_{t+1}]/(n-1)$

S Sequence evolution

S1 First-seen tools

mean $\mathbf{1}[c_t \notin \mathcal{H}_{t-1}]$

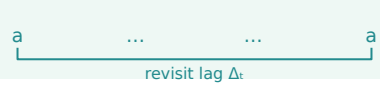


S2 Revisit share

$\Pr(\Delta_t \geq 2 \mid \text{repeated})$

S3 Revisit lag

mean $\{\log(1+\Delta_t)/\log 129\}$

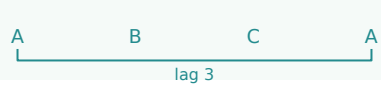


S4 A–B–A returns

$\Pr(a_t = a_{t+2} \mid a_t \neq a_{t+1})$

S5 Lag-three returns

$\Pr(a_t = a_{t+3})$



S6 Bigram reuse

mean $\mathbf{1}[(c_{t-1}, c_t) \in \mathcal{H}_2]$

S7 Trigram reuse

mean $\mathbf{1}[c_{t-2:t} \in \mathcal{H}_3]$

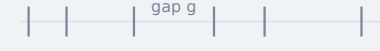
S8 Window distribution drift

$\mathrm{JSD}(p_j, p_{j-1})$

T Observed timing

T1 Typical log gap

median(ℓ)



T2 Gap interquartile range

$Q_{.75}(\ell) - Q_{.25}(\ell)$

T3 Burstiness

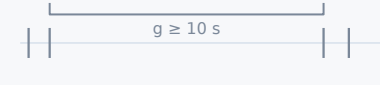
$(\sigma_g - \mu_g)/(\sigma_g + \mu_g)$

T4 Fast-gap share

mean $\mathbf{1}[g \leq 1\,\text{s}]$

T5 Long-pause share

mean $\mathbf{1}[g \geq 10\,\text{s}]$



T6 Serial gap dependence

$\mathrm{corr}(\ell_t, \ell_{t+1})$

T7 Execution duration

median $\log(1+d)$

T8 Idle after completion

median $\log(1+i)$